MATTIA TARABOLO

Torino, 10125
m a t t i a [dot] t a r a b o l o [at] g m a i l [dot] c o m
0000-0003-1569-5223

EDUCATION

PhD in Physics| Politecnico di Torino (Torino, Italy)
NOVEMBER 2022 – JANUARY 2026

- Thesis title: "Cavity-Based Approaches to Stochastic Dynamics on Sparse Graphs"
- Working on stochastic systems on sparse graphs using statistical physics' inspired methods.
- Member of the Board of the PhD in Physics as PhD students' representative.

Master of Science programme in Physics of Complex Systems| Université Paris Cité (Paris, France)
SEPTEMBER 2021 – JULY 2022

- Mark: 17.6/20, Très Bien.
- Thesis title: "Bayesian variable selection for environment-dependent phylogenetic models of diversification".
- Ranked 1st out of ~ 60 students in the first semester, with an average of 18.6.
- Course taught entirely in English.

Master of Science programme in Physics of Complex Systems| Politecnico di Torino (Turin, Italy)
SEPTEMBER 2020 – OCTOBER 2022

- Mark: 110/110 Cum Laude.
- Thesis title: "Bayesian variable selection for environment-dependent phylogenetic models of diversification".
- Admitted to the international track.
- Attended the first semester at SISSA and ICTP (Trieste, Italy), the second semester at Politecnico di Torino (Torino, Italy) and the third semester at Sorbonne Université, Université de Paris and Université Paris Saclay (Paris, France).
- Course taught entirely in English.

Bachelor of Science in Physics | Università degli Studi dell'Insubria (Como, Italy)
SEPTEMBER 2017 – OCTOBER 2020

- Mark: 110/110 cum laude.
- Thesis title: "Generation of prime numbers from a random bit stream".
- Scholarship of excellence and merit.

SCIENTIFIC INTERESTS

- Applications of statistical physics to optimization and inference problems, with a focus on stochastic processes on graphs. Strong background in mean field methods and message passing algorithms.
- Non-equilibrium stochastic systems on graphs. Experienced in working with path integral techniques for the study of stochastic processes. Currently studying the application of dynamic cavity methods to study the dynamics of Stochastic Differential Equations (SDEs) on diluted graphs, with a focus on complex ecosystems and neural networks.
- Random matrices. Experienced in cavity methods for the study of non-hermitian sparse random matrices.
- Interested in spin-glass theory, applications of statistical physics to neuroscience, machine learning.

TEACHING

- Senior tutor of the Physics I course at Politecnico di Torino (AY 2023/24 and AY 2024/25).
- Consolidated experience in tutoring students from elementary school to high school.

ACTIVITIES

- **Talk accepted at "Decoding Complexity in Ecology and Economics: Statistical Physics & Data-Driven approaches" Workshop**, University of Cagliari, Cagliari, July 2025.
- **Talk accepted at "Biological Physics and Statistical Mechanics" Conference"**, University of Naples Federico II, Naples, June 2025.
- **Poster accepted at "Stochastics processes: Inferences in complex systems" CECAM Workshop**, CECAM-HQ-EPFL, Lausanne, May 2025.
- **Visiting at Center of Complex Systems, Physics Faculty of the University of Havana.** Worked with Prof. Roberto Mulet on sparse disordered complex ecosystems using statistical physics tools.
- **Beg Rohu Summer School of physics "Concepts and Methods of Statistical Physics"**, Saint-Pierre Quiberon, June 2024.
- **Talk accepted at "Complex networks: from socio-economic systems to biology and the brain" Workshop**, Lipari, July 2023.
- **Beg Rohu Summer School of physics "Statistical Physics of Complex Systems"**, Saint-Pierre Quiberon, June 2023.
- **CAMBI (Computational Aspects and Modeling of Biological Information) Workshop**, Bocconi University, Milan, December 2022.
- **Spring College in the Physics of Complex Systems (smr 3690)**, ICTP, Trieste, February/March 2022.
- **Winter School on Quantitative Systems Biology: Quantitative Approaches in Ecosystem Ecology (smr 3486)**, ICTP, Online, November/December 2020.

PROJECTS

- **SIMBAD - Statistical Inference from Multiscale Biological Data: theory, algorithms, applications**, (2023-2027) UE-funded research - HE - Excellent Science – MSCA

PUBLICATIONS

- Braunstein, G. Catania, L. Dall'Asta, M. Mariani, F. Mazza and **M. Tarabolo**, *Small-Coupling Dynamic Cavity: A Bayesian mean-field framework for epidemic inference*, Phys. Rev. Res. 7, 023089 (2025).
- **M. Tarabolo** and L. Dall'Asta, *Gaussian approximation of dynamic cavity equations for linearly-coupled stochastic dynamics*, SciPost Phys. 19, 019 (2025).
- **M. Tarabolo**, L. Dall'Asta, R. Mulet, *Sparse Interactions Reshape Stability in Random Lotka-Volterra Dynamics*, arXiv:2512.22563 (2025).

SKILLS

Language

- **Italian**: mother tongue.
- **English**: C1 level (IELTS Academic test, mark 7.5).
- **French**: elementary level.

IT

- **Programming**: advanced user of Matlab, Python, Julia. Intermediate user of C++ and R.
- **Operating system**: advanced user of Windows and Unix-based systems.
- **Graphics and editing:** intermediate user of Photoshop, Lightroom, Gimp, InkScape, Vegas Pro.